

Milad Tatar Mamaghani

Research Scientist | 6G Wireless Systems, Signal Processing & Applied AI

Sydney, NSW, Australia

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🏠 Homepage • 📄 Google Scholar • 🔗 LinkedIn • 📄 ORCID • 🐙 GitHub

PROFESSIONAL PROFILE

Research scientist and engineer with a PhD in Electrical Engineering and 8+ years of research & development experience spanning wireless 6G networks, statistical signal processing, optimisation, and applied machine learning. Author of 17+ publications in leading IEEE venues (h-index 13 and nearly 700 citations), and recipient of two IEEE best paper awards on communications and signal processing, with deep expertise in physical-layer security, wireless UAV communications, and integrated sensing and communications. Strong background in mathematical modelling, algorithm development, reinforcement learning, and modern software engineering to solve complex technical challenges.

RESEARCH INTERESTS

My research develops analytical and learning-based methods for secure, reliable, energy-efficient, and resilient next-generation wireless systems. I combine tools across information theory, signal processing, mathematical optimisation, and reinforcement learning to design adaptive resource-allocation and control strategies, enabling various 6G applications.

- AI-native wireless communication networks
- Physical-layer security, privacy, and network resilience
- Integrated sensing, communications, and computing
- Machine-type communications and URLLC
- UAV communications and non-terrestrial networks
- Delay-Doppler signal processing for passive radar

RESEARCH & PROFESSIONAL EXPERIENCE

Technical Subject-Matter Expert Consultant & Researcher

Feb. 2025–Present

Mercor AI (Remote)

Sydney, Australia

- Designed complex domain-specific LLM training and evaluation datasets across mathematics, coding, and engineering for top-tier AI labs.
- Developed structured review and evaluation workflows, surpassing a 95% quality rate, to improve model reasoning capability.
- Co-authored 2+ journal papers on NOMA-ISAC, physical-layer anonymous communications, and delay-Doppler signal processing for ISAC.

ARC Postdoctoral Research Fellow — Secure Machine-Type Communications

Jan. 2023–Jan. 2025

The Australian National University, School of Engineering

Canberra, Australia

- Derived information-theoretic performance metrics for average information leakage in finite-blocklength wiretap fading channels.
- Designed joint resource allocation and 3D UAV-trajectory algorithms for secure short-packet aerial relaying systems under propulsion limits.
- Formulated a game model for ISAC under mobile eavesdropping and developed DRL agents for adaptive trajectory and beamforming control.
- Produced three first-author Q1 IEEE journal articles (1x JSAC, 2x TWC) and two conference papers (IEEE ICC 2024, IEEE GLOBECOM 2023).
- Received two best-paper awards (IEEE SPCC-TC BPA 2024; IEEE ICC BPA 2024 — top 0.6% of 2,364 submissions).

Doctoral Researcher — Wireless UAV Communications

Jan. 2019–Dec. 2022

Monash University, Department of Electrical and Computer Systems Engineering

Melbourne, Australia

- Designed reinforcement learning and mathematical optimisation frameworks for energy-efficient, secure airborne communications.
- Formulated and solved problems on aerial beamforming, spatial jamming, and trajectory planning to secure UAV network architectures.
- Proposed learning-based system designs for secure MIMO-UAV relaying, achieving significant performance improvements over baselines.
- Developed reproducible MATLAB/Python simulations to validate analytical secrecy and reliability models under practical constraints.
- Published seven peer-reviewed articles across leading journals; received several awards, grants, and accolades for research excellence.

Data Quality & Automation Analyst

Oct. 2022–Dec. 2022

Monash University, Office of the Deputy Vice-Chancellor

Melbourne, Australia

- Built Python ETL pipelines integrating a large number of research records from heterogeneous data sources into a unified schema.
- Developed reusable data-quality and reporting tools for automated validation, normalisation and reconciliation checks.

Research Assistant — Untrusted Wireless-Powered Relaying

Jan. 2016–Nov. 2018

Amirkabir University of Technology, Wireless Communication Research Lab

Tehran, Iran

- Derived secrecy outage probability and ergodic secrecy rate for wireless-powered amplify-and-forward untrusted relay networks.
- Designed cooperative-jamming and wireless-energy-harvesting schemes for secure multi-hop relaying scenarios with sensitivity analysis.
- Produced two journal articles and one conference paper on secure IoT networks via simultaneous wireless information and power transfer.

Summer Research Intern

Jun. 2016–Sep. 2016

Iran Telecommunication Research Center (ITRC) — Fixed Communications Group

Tehran, Iran

- Worked on healthcare IoT networks; gained hands-on experience in applied telecommunications and network infrastructure.

EDUCATION

Doctor of Philosophy (PhD) in Electrical Engineering

Monash University - Clayton Campus

Jan. 2020–Dec. 2022

Melbourne, Australia

- Thesis: *Safeguarding Beyond-5G Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization*
- Pathway: Transferred from Master of Engineering Science (Research) to PhD (Jan. 2019–Dec. 2020)

Bachelor of Science (BSc) in Electrical–Communications Engineering

Amirkabir University of Technology (Tehran Polytechnic)

Sep. 2012–Oct. 2016

Tehran, Iran

- Thesis: *Secure Communications with Untrusted Relaying and Wireless Energy Harvesting: Performance Analysis and Trade-offs*
- Double degree: BSc in Control Engineering with extra course credits (Feb. 2018)

SELECTED RESEARCH PROJECTS

Integrated Sensing and Communications for 6G

2025–Present

Security, AI-Driven Resource Allocation, Stackelberg Games, Deep Reinforcement Learning

- Developed the first Stackelberg-game framework for securing UAV-enabled ISAC against a mobile adversary, jointly modelling sensing, communications and eavesdropping. Designed and benchmarked deep-reinforcement-learning agents for adaptive trajectory and beamforming control, enabling autonomous protection of dual-function 6G infrastructure as threats and channel conditions evolve.

Physical-Layer Security for Wireless Machine-Type Communications

2023–2025

Finite Blocklength Communications, IoT, URLLC, Information-Theoretic Security

- Led ARC-funded research on secure short-packet transmission for IoT and mission-critical networks. Derived and validated the first analytical framework for average information leakage over finite-blocklength fading channels, and designed joint blocklength, resource-allocation and UAV-trajectory algorithms that expose actionable security–latency–reliability trade-offs.

Safeguarding Wireless UAV Communications

2019–2022

UAV Networks, Trajectory Design, THz Communications, Learning-Based Control

- Designed optimisation and model-free reinforcement-learning methods for secure, reliable, and energy-efficient UAV communications, spanning trajectory control, artificial-noise allocation, MIMO beamforming design, and aerial intelligent reflecting surfaces to enable confidential and resilient airborne connectivity under mobility and propulsion constraints.

Wireless-Powered Untrusted Relaying Networks

2016–2021

Wireless Energy Harvesting, Cooperative Jamming, Physical-Layer Security

- Derived secrecy-outage and ergodic-secrecy models for amplify-and-forward networks in which energy-constrained relays cannot be trusted. Designed wireless-energy-harvesting and cooperative-jamming schemes that allow relays to forward sensitive data without gaining useful information, supporting secure and energy-efficient cooperative coverage in IoT networks.

SELECTED APPLIED PROJECTS

InvestIQ — Multi-Asset Portfolio Analytics Platform

Python, FastAPI, PostgreSQL, Redis, Next.js, TypeScript, Docker

- Engineered an event-sourced portfolio platform that consolidates multi-broker holdings across 15+ asset classes, normalises foreign-currency positions and automates reconciliation and lot-level CGT tracking. Added performance and risk analytics plus an LLM-assisted adviser, giving investors a unified, tax-aware view of diversified portfolios and decision-ready insights. [GitHub]

Healthmate — Mobile-First Wellness Progressive Web App

Next.js, TypeScript, React, Tailwind CSS, Supabase, Anthropic API, Vitest

- Built a local-first wellness PWA for Mediterranean meal planning, grocery generation, habit and body-measurement tracking, and progress visualisation. Added deterministic meal scoring and safety-constrained coaching with an offline fallback, enabling private day-to-day guidance without mandatory cloud services. [GitHub]

TradingAgents — Multi-Agent LLM Investment Research Platform

Python, LangGraph, FastAPI, React, TypeScript

- Turned an open-source multi-agent LLM framework into an interactive investment-research platform by adding a web console, persistent analysis runs, configurable models and strategy visualisation. The resulting workflow makes agent reasoning, evidence and recommendations easier to inspect, compare and reproduce before investment decisions are made. [GitHub]

CNN Image Classification Benchmark

Python, TensorFlow, Keras, NumPy

- Built a reproducible image-classification pipeline to train and evaluate baseline CNNs, data augmentation and transfer-learning approaches under a consistent benchmark. Achieved 96.0% test accuracy and produced a reusable experimental workflow for selecting accurate models and quantifying the benefit of each training strategy. [GitHub]

HONOURS & AWARDS

- 2025 **IEEE ComSoc Best Readings in Covert Communication and Networking** [\[Link\]](#)
 - Recognises impactful and landmark technical contributions in the field (featured paper [9]).
- 2024 **IEEE ICC Best Paper Award** [\[Certification\]](#)
 - Selected in the top 0.6% of 2,364 submissions across all symposia.
- 2024 **IEEE Signal Processing and Computing for Communications Best Paper Award** [\[Certification\]](#)
 - Recognises outstanding contributions to the field by the Technical Committee; only one paper is awarded each year.
- 2024 **Australian Global Talent**
 - Recognises individuals of exceptional achievement in technology under the national innovation program.
- 2023 **Research Fellowship Offer, KTH Royal Institute of Technology**
 - Selected from 60+ international applicants; offer declined due to personal and career alignment reasons.
- 2022 **Exceptional PhD Thesis**
 - Recognised among the top 5% of PhD theses in the field by external examiners.
- 2019 **Vice-Chancellor's Congratulatory Letter for Teaching Excellence**
 - Awarded for outstanding performance in teaching and learning based on student evaluations at Monash University.
- 2014 **Exceptional Talent Recognition, Amirkabir University of Technology**
 - Recognised for outstanding academic excellence during undergraduate studies at Amirkabir University of Technology.
- 2012 **Top 0.2%, National University Entrance Examination (Konkour)**
 - Ranked in the top 0.2% among nearly 300,000 applicants nationwide.
- 2011 **Semifinalist, National Mathematics and Chemistry Olympiads**
 - Selected among top performers in national mathematics and chemistry competitions.

GRANTS & FUNDING

- 2023 **ANU Early Career Researcher International Travel Grant (AUD 3000)**
 - Awarded by the Australian National University for presenting research at international conferences.
- 2022 **Monash Postgraduate Publications Award (AUD 5000)**
 - Awarded for outstanding research publications during doctoral candidature.
- 2022 **Monash Graduate Research Completion Award**
 - Awarded for successful and timely completion of doctoral research.
- 2022 **Monash Engineering Faculty's Postgraduate Research Scholarship**
 - Awarded for outstanding research potential and academic achievement in the field of engineering.
- 2019 **Monash Graduate Scholarship and International Tuition Scholarship**
 - Competitive scholarship, including a stipend (AUD 120,000) and a full tuition waiver (AUD 200,000) for the doctoral studies.
- 2019 **Melbourne Research Scholarship | Macquarie Research Excellence Scholarship | International Silver Scholarship**
 - Merit-based postgraduate scholarship offers from the University of Melbourne, Macquarie University, and Politecnico di Milano.

TECHNICAL SKILLS

Programming	Python, MATLAB, C/C++, SQL (T-SQL, PostgreSQL, MySQL), NumPy, SciPy, Matplotlib, Pandas, VHDL
Signal Processing	Statistical modelling, detection & estimation, filtering, spectral analysis, hypothesis testing
Wireless Systems	5G/NR, 3GPP, IEEE 802.11, OFDM, MIMO, beamforming, channel modelling, GNU Radio, SDR
Optimisation	convex programming, nonlinear programming, stochastic optimisation, CVX, MOSEK, Gurobi
Machine Learning	PyTorch, TensorFlow, scikit-learn, recommenders, classification, clustering
Deep Learning & AI	Neural Networks (CNNs, LSTMs), deep reinforcement learning, Transformers, NLP, Computer Vision
Generative AI & LLM	LangChain, LangGraph, RAG, prompt engineering, Anthropic/OpenAI APIs, agentic workflows
Software & DevOps	Docker, Azure, Linux, Bash, Git/GitHub, CI/CD, pytest, FastAPI, REST APIs, web scraping

PROFESSIONAL DEVELOPMENT

- **Data Scientist Professional** [\[Certificate\]](#) — DataCamp, Aug. 2026.
- **Deep Learning Specialization** [\[Certificate\]](#) — DeepLearning.AI, Oct. 2022.
- **Machine Learning Specialization** [\[Certificate\]](#) — Stanford Online, Apr. 2020.

TEACHING EXPERIENCE

Teaching Associate

Monash University, Faculty of Engineering

Feb. 2019–Dec. 2022

Melbourne, Australia

- Delivered tutorials, laboratories, and practical sessions across 14 university courses, supporting cohorts of 500+ students each semester.
- Contributed to curriculum, designed course materials and assessments, coordinated examinations and projects, and mentored students.
- Recognised for teaching excellence; achieved a student satisfaction score of 4.73/5.0, ranking within the top 7.3% of Monash units.

Courses Taught

- ECE5179: Neural Networks and Deep Learning (Postgraduate/Undergraduate); Tutor and Lab Demonstrator, Semester 2, 2022
- ECE5883: Advanced Signal Processing (Postgraduate); Lab Designer and Demonstrator, Semester 1, 2022
- ENG5001: Advanced Engineering Data Analysis (Postgraduate/Undergraduate); Tutor, Semester 1, 2022
- FIT1048: Fundamentals of C++ (Undergraduate); Tutor and Lab Demonstrator, Semester 1, 2022
- ECE5884: Wireless Communications (Postgraduate); Lab Demonstrator, Semester 2, 2021
- ECE4191: Engineering Design (Undergraduate); Lab Demonstrator and Supervisor, Semester 2, 2020–2022
- ENG5105: Integrated Design (Postgraduate); Coordinator and Project Mentor, Semester 2, 2020–2022
- ECE4146: Multimedia Technologies (Postgraduate/Undergraduate); Lab Demonstrator, Semester 2, 2020–2022
- ECE3141: Information and Networks (Undergraduate); Lab Demonstrator, Semester 1, 2020–2022
- ECE4044: Telecommunications Protocols (Undergraduate); Lab Demonstrator, Semester 1, 2020
- ECE2072: Digital Systems (Undergraduate); Tutor and Lab Demonstrator, Semester 1, 2020–2022
- ECE4042: Communications Theory (Undergraduate); Tutor and Lab Demonstrator, Semester 1, 2019–2022
- ECE4076: Computer Vision (Postgraduate/Undergraduate); Tutor, Lab Demonstrator, and Coordinator, Semester 1, 2019–2022
- ECE2071: Computer Organisation and Programming (Undergraduate); Tutor and Lab Demonstrator, Semester 1, 2019–2022

Teaching Assistant

Amirkabir University of Technology, Department of Electrical Engineering

Jan. 2016–Oct. 2017

Tehran, Iran

- Contributed to Communications I and Computer Programming courses, guiding students through practical experiments and coding exercises.

SUPERVISION & MENTORSHIP

Research Supervision — Australian National University

2023–2026

Co-supervised one PhD candidate, one Engineering R&D Honours student, and one visiting undergraduate student.

Yu Li — PhD Candidate, Australian National University, 2023–2026

- Topic: Physical-Layer Anonymous Communications; co-supervised with A/Prof. Xiangyun Zhou.
- Output: One IEEE journal submission and one manuscript in preparation.

Mac Gilligan — Engineering R&D Honours Student, Australian National University, 2024–2025

- Topic: Software-defined radio implementation of an over-the-air file transmission with GNU Radio.
- Output: ANU University Medal nominee for academic excellence.

Reetika — Visiting Undergraduate Student, Indian Institute of Space Science and Technology, 2023–2024

- Topic: Physical-layer security for terahertz communications; co-supervised with Prof. Nan Yang.

Engineering Design Project Supervision — Faculty of Engineering, Monash University

2021–2022

Supervised Undergraduate and Postgraduate Capstone Projects

- **Automated Robotic Arm:** Coloured-cube sorting robotic arm with microprocessor-based controller, 2021.
- **Autonomous Mobile Robot:** Odometric sensing for high-level navigation and autonomous ball collection, 2022.
- **Intelligent Baby Monitor:** Sustainable monitor with voice/image recognition and real-time parent alerts, 2022.

SERVICE & LEADERSHIP

Conference Organisation and Service

- **Technical Program Committee Member** — IEEE Vehicular Technology Conference, 2024.
- **Session Chair** — IEEE Global Communications Conference, Physical Layer Security Symposium, 2023.
- **Selection Committee Member** — ANU Summer Research Scholarship Program, 2023.
- **Assessment Team Member** — Monash University Online Exams & Student Services, 2021–2023.

Peer-Review Contributions

- **Volunteer Technical Reviewer** — Over 145 verified reviews for 25+ international journals and conferences, 2018–Present.
 - **Journals:** IEEE JSAC, TWC, TCOM, TIFS, TSP, TVT, TCCN, TMC, TGCN, TNSE, IoT-J, Commun. Lett., WCL, Commun. Mag., and others.
 - **Conferences:** IEEE ICC (2020–2023), IEEE GLOBECOM (2017, 2024–2026), IEEE VTC (2024), WiOpt (2023), WCSP (2023), CyberC (2020).

Professional Memberships

- IEEE Member (2022–2026); IEEE Graduate Student Member (2020–2022).
- IEEE Communications Society (2020–2025); IEEE Signal Processing Society (2020–2022); IEEE Young Professionals (2020–2025).

SELECTED TALKS & PRESENTATIONS

On the Information Leakage Performance of Secure Finite Blocklength Transmissions [Slides] [Certificate]

Jun. 2024

IEEE ICC 2024 — Communication Theory Symposium: Physical-Layer Security and Privacy — Denver, USA

Secrecy Performance of Finite Blocklength Communications over Fading Channels [Poster]

Feb. 2024

21st Australian Communications Theory Workshop (AusCTW) — Melbourne, Australia

Short-Packet Transmissions with Aerial Relaying: Blocklength and Trajectory Co-Design [Slides] [Certificate]

Dec. 2023

IEEE GLOBECOM 2023 — Communication and Information System Security Symposium — Kuala Lumpur, Malaysia

Safeguarding Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization [Slides]

Feb. 2022

PhD Final Review Seminar — Department of ECSE, Monash University — Melbourne, Australia

Performance Analysis of Secure Communications via Untrusted Relaying with Wireless Energy Harvesting

Sep. 2016

BSc Thesis Defence [in Persian] — Department of Electrical Engineering, Amirkabir University of Technology — Tehran, Iran

PUBLICATIONS

- **Research areas:** Wireless Communications, Signal Processing, Information Theory, Optimisation, and Machine Learning
- **Scholarly outputs:** 12 journal articles, 3 conference papers, 2 preprints, and 2 theses
- **Research impact:** 696 citations, h-index 13, i10-index 13 (as of September 2026)

Journals

- [12] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, "Securing integrated sensing and communication against a mobile adversary: A Stackelberg game with deep reinforcement learning," *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 942-958, Sep. 2025. (IF 17.2, **Q1**)
DOI 10.1109/JSAC.2025.3611404 |  PDF |  arXiv |  Code
- [11] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, Lee Swindlehurst, and Vincent Poor, "Performance analysis of finite blocklength transmissions over wiretap fading channels: An average information leakage perspective," *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 13252-13266, Oct. 2024. (IF 10.4, **Q1**)
DOI 10.1109/TWC.2024.3400601 |  PDF |  arXiv |  Code
- [10] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, "Secure short-packet communications via UAV-enabled mobile relaying: Joint resource optimization and 3D trajectory design," *IEEE Transactions on Wireless Communications*, vol. 23, no. 7, pp. 7802-7815, July 2024. (IF 10.4, **Q1**)
DOI 10.1109/TWC.2023.3344802 |  PDF |  arXiv |  Code
- [9] **Milad Tatar Mamaghani** and Yi Hong, "Aerial intelligent reflecting surface-enabled terahertz covert communications in beyond-5G Internet of Things," *IEEE Internet of Things Journal*, vol. 9, no. 19, pp. 19012-19033, March 2022. (IF 9.9, **Q1**)  **IEEE ComSoc Best Readings**
DOI 10.1109/JIOT.2022.3163396 |  PDF |  arXiv |  Code
- [8] **Milad Tatar Mamaghani** and Yi Hong, "Terahertz meets untrusted UAV-relaying: Minimum secrecy energy efficiency maximization via trajectory and communication co-design," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 5, pp. 4991-5006, May 2022. (IF 6.41, **Q1**)
DOI 10.1109/TVT.2022.3150011 |  PDF |  arXiv
- [7] **Milad Tatar Mamaghani** and Yi Hong, "Joint trajectory and power allocation design for secure artificial noise aided UAV communications," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 3, pp. 2850-2855, Mar. 2021. (IF 6.41, **Q1**)
DOI 10.1109/TVT.2021.3057397 |  PDF |  arXiv
- [6] **Milad Tatar Mamaghani** and Yi Hong, "Intelligent trajectory design for secure full-duplex MIMO-UAV relaying against active eavesdroppers: A model-free reinforcement learning approach," *IEEE Access*, vol. 9, pp. 4447-4465, Dec. 2020.
DOI 10.1109/ACCESS.2020.3048021 |  PDF
- [5] **Milad Tatar Mamaghani** and Yi Hong, "Improving PHY-security of UAV-enabled transmission with wireless energy harvesting: Robust trajectory design and communications resource allocation," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 8, pp. 8586-8600, May 2020. (IF 6.41, **Q1**)
DOI 10.1109/TVT.2020.2998060 |  PDF |  arXiv
- [4] **Milad Tatar Mamaghani**, Ali Kuhestani, and Hamid Behrouzi, "Can a multi-hop link relying on untrusted amplify-and-forward relays render security?," *Wireless Networks*, vol. 27, pp. 795-807, Nov. 2020.
DOI 10.1007/s11276-020-02487-w |  PDF |  arXiv
- [3] **Milad Tatar Mamaghani** and Yi Hong, "On the performance of low-altitude UAV-enabled secure AF relaying with cooperative jamming and SWIPT," *IEEE Access*, vol. 7, pp. 153060-153073, Oct. 2019.
DOI 10.1109/ACCESS.2019.2948384 |  PDF |  Code
- [2] **Milad Tatar Mamaghani** and Robert Abbas, "Security and reliability performance analysis for two-way wireless energy harvesting based untrusted relaying with cooperative jamming," *IET Communications*, vol. 13, no. 4, pp. 449-459, Mar. 2019.
DOI 10.1049/iet-com.2018.5718 |  PDF
- [1] **Milad Tatar Mamaghani**, Ali Kuhestani, and Kai Kit Wong, "Secure two-way transmission via wireless-powered untrusted relay and external jammer," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 9, pp. 8451-8465, July 2018. (IF 6.41, **Q1**)
DOI 10.1109/TVT.2018.2848648 |  PDF |  arXiv

Conferences

- [3] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, Lee Swindlehurst, and Vincent Poor, “On the average information leakage of finite blocklength transmissions over Rayleigh fading channels,” in *Proc. IEEE International Conference on Communications*, Denver, CO, USA, pp. 1467-1472, June 2024. 🏆 **IEEE Best Paper Award**
DOI 10.1109/ICC51166.2024.10622223 | [arXiv](#)
- [2] **Milad Tatar Mamaghani**, Xiangyun Zhou, Nan Yang, and Lee Swindlehurst, “Secure short-packet transmission with aerial relaying: Blocklength and trajectory co-design,” in *Proc. IEEE Global Communications Conference*, Kuala Lumpur, Malaysia, pp. 5998-6004, Dec. 2023.
DOI 10.1109/GLOBECOM54140.2023.10437972 | [arXiv](#)
- [1] **Milad Tatar Mamaghani**, Abbas Mohammadi, Phil Yeoh, and Ali Kuhestani, “Secure two-way communication via a wireless-powered untrusted relay and friendly jammer,” in *Proc. IEEE Global Communications Conference*, Singapore, pp. 1-6, Dec. 2017.
DOI 10.1109/GLOCOM.2017.8253999 | [arXiv](#)

Under Review

- [2] Azadeh Tabeshnezhad, **Milad Tatar Mamaghani**, Lee Swindlehurst, Tommy Svensson, and Erik Ström, “Robust Beamforming Design for Secure Uplink NOMA-ISAC,” submitted to *IEEE Transactions on Wireless Communications*, Jun. 2026.
DOI 10.48550/arXiv.2606.17306 | [arXiv](#)
- [1] Yu Li, **Milad Tatar Mamaghani**, Xiangyun Zhou, and Nan Yang, “Symbol-Aware Precoder Design for Physical-Layer Anonymous Communications,” submitted to *IEEE Transactions on Communications*, Mar. 2026.
DOI 10.48550/arXiv.2602.20874 | [arXiv](#)

Theses

- [2] **Milad Tatar Mamaghani**, “Safeguarding Beyond-5G Wireless Communications with Unmanned Aerial Vehicles: Design and Optimization,” PhD Thesis, Electrical and Computer Systems Engineering Department, Monash University, Melbourne, Australia, Dec. 2022.
DOI 10.26180/21721709.v1
- [1] **Milad Tatar Mamaghani**, “Secure Communications via Untrusted Relaying with Wireless Energy Harvesting: Performance Analysis and Trade-offs,” *BSc Thesis* [in Persian], Department of Electrical Engineering, Amirkabir University of Technology, Tehran, Iran, Sep. 2016.

LANGUAGES

English (Professional fluency) | Persian/Farsi (Native) | Azerbaijani (Native) | Turkish (Intermediate)

REFERENCES

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| • Xiangyun Zhou — Associate Professor, Australian National University, Australia. | xiangyun.zhou@anu.edu.au |
| • Lee Swindlehurst — Distinguished Professor and Department Chair, UC Irvine, USA. | swindle@uci.edu |
| • Yi Hong — Associate Professor, Monash University, Australia. | yi.hong@monash.edu |
| • Nan Yang — Professor & Research Cluster Lead, Australian National University, Australia. | nan.yang@anu.edu.au |
| • Vincent Poor — Michael Henry Strater Professor, Princeton University, USA. | poor@princeton.edu |
| • Abbas Mohammadi — Professor, Amirkabir University of Technology, Iran. | abm125@aut.ac.ir |
| • Ali Kuhestani — Assistant Professor, Qom University of Technology, Iran. | kuhestani@qut.ac.ir |